

A Banach-Envelope Obstruction for Separably Injective p -Banach Spaces

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Source Question

The source paper is Cabello Sánchez, Garbulińska-Węgrzyn, and Kubiś, *Quasi-Banach spaces of almost universal disposition*, arXiv:1309.7649. In Section 7.4, after proving that $c_0(E)$ is separably injective whenever E is separably injective among p -Banach spaces, the authors ask whether there is a nontrivial separable space separably injective among p -Banach spaces when $p < 1$, and say that their guess is negative.

Definitions

Let $0 < p \leq 1$. A p -Banach space is a complete quasi-normed space whose quasi-norm satisfies

$$\|x + y\|^p \leq \|x\|^p + \|y\|^p.$$

Such a space E is separably injective among p -Banach spaces if, for every separable p -Banach space X , every subspace $Y \subset X$, and every bounded operator $t : Y \rightarrow E$, there is a bounded extension $T : X \rightarrow E$.

Partial Result

Theorem 1. *Let $0 < p < 1$, and let E be separably injective among p -Banach spaces. If F is a q -Banach space for some $p < q \leq 1$, then every bounded linear operator $S : E \rightarrow F$ is zero.*

Proof Intuition

Separably injectivity lets one extend maps out of one-dimensional subspaces of separable p -Banach spaces. Apply it to the constants inside $L_p[0, 1]$. If E had any nonzero operator into a more convex q -Banach target, then the extension would produce a nonzero operator $L_p[0, 1] \rightarrow F$. Kalton's classical theorem forbids exactly such operators when $p < q \leq 1$.

Proof

Assume, toward a contradiction, that $S : E \rightarrow F$ is nonzero. Choose $e \in E$ with $S(e) \neq 0$. Let $L_p = L_p[0, 1]$, with its usual p -norm, and let $Y \subset L_p$ be the one-dimensional subspace of constant

functions. Since the measure of $[0, 1]$ is one, the map

$$t : Y \rightarrow E, \quad t(\lambda \mathbf{1}) = \lambda e$$

is bounded.

The space $L_p[0, 1]$ is separable and is a p -Banach space. By separable injectivity of E , the operator t extends to a bounded linear operator

$$T : L_p[0, 1] \rightarrow E.$$

Then $S \circ T : L_p[0, 1] \rightarrow F$ is a bounded linear operator and

$$(S \circ T)(\mathbf{1}) = S(e) \neq 0.$$

This contradicts the classical result, quoted in the source paper in Section 7.5, that every operator from L_p to a q -Banach space is zero whenever $p < q \leq 1$. Therefore $S = 0$.

Corollary 1. *If $0 < p < 1$ and E is separably injective among p -Banach spaces, then $E^* = \{0\}$. Moreover, the Banach envelope of E is trivial.*

Proof. The scalar field $\mathbb{K}$ is a 1-Banach space, so the theorem gives $E^* = \{0\}$. The canonical map from E into its Banach envelope is a bounded operator from E into a Banach space. Applying the theorem with $q = 1$, this canonical map must be zero. Hence the Banach envelope is the zero space. $\square$

Scope and Limitations

This is only a partial obstruction to the source question. It does not rule out a nontrivial separable separably injective p -Banach space. It shows that any such example must have no nonzero maps into Banach spaces, no nonzero dual, and no nonzero Banach envelope. In particular, locally convex examples and examples detectable by Banach-envelope methods are impossible.

Novelty Check

The lightweight run indexes were searched for arXiv:1309.7649, separably injective p -Banach spaces, Banach envelope, and the L_p -to- q -Banach zero-operator obstruction. No existing packet or attempt for this partial obstruction was found. Web searches for later work on p -Gurarii spaces, separably injective quasi-Banach spaces, and the K -space/quotient questions surfaced the source paper but no later exact answer.

References

- F. Cabello Sánchez, J. Garbulińska-Węgrzyn, and W. Kubiś, *Quasi-Banach spaces of almost universal disposition*, arXiv:1309.7649.
- N. J. Kalton, *Locally complemented subspaces and $\mathcal{L}_p$ -spaces for $p < 1$* , Math. Nachr. 115 (1984), 71–97. The zero-operator theorem used above is quoted in Section 7.5 of the source paper.

Human Review Recommendation

This packet is short and should be checked mainly for the exact formulation of the Kalton zero-operator theorem and for the convention that $L_p[0, 1]$ is included among the p -Banach spaces allowed in the separable-injectivity definition. Under those standard conventions, the proof is immediate.